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My Personal Profile

Academic Info And Background

My name is **Muhammad Huzaifa Ayub**. I am 19 years Old. Currently, I am doing *Bachelor's* in **Data Science** from **PUCIT (*Punjab University College of Information and Technology*)** , which is one of *Top-rated* and *prestigious* university of **Pakistan**. As of now , I am in ***Third Semester***. I live in **Lahore** , which is also known as *Heart* of Pakistan.

Interests And Goal

Since my childhood, I had a keen curiosity of **How things Works** , like ***How a plane maintain a high-speed in air ? How a Car Engine converts gasoline into Mechanical Power ? How websites are Built ? How a small CPU can perform Billions of operations/second ? How a pistol can shoot a small bullet at several Km/s ? How radar detects anything ? How Quantum Chips computes ?***

I know these are some obvious and common questions. But these questions means a lot to me , because after knowing how these things works , I started thinking differently and logically. My curiosity for knowing things changes *time-to-time*. But I have n't quit learning .

> In my opinion , *learning is a great Opportunity.*

After Exploring different things, I found myself as ***Computer Nerd.*** Therefore, I decided to pursue my career in Computer Science , more specifically in Data Science.I like to solve problems in which we have to works on different things to get our (useful for us) result. I love to Solve problems in which different things works together to form a big Result/Combined Effect.

I am still learning , still exploring, I hope my decision will turn out to be a excellent future career.

Copyable Quick Info (JSON)


```
{
    "Name" : "Muhammad Huzaifa Ayub" ,
    "Age" : 19 ,
    "University" : "PUCIT",
    "Semester" : 3 ,
    "Matric_percentage" : 96.3 ,
    "Intermediate_percentage" : 88.8 ,
    "Hobbies" : [ "Gaming" , "Jail-Breaking things" , "Vibe-Coding" , "Astronomy" ,
"Benchmarking PC Components & Codes" , "Watching Movies depicting Sci-fi" , "Solving
logical problems"]
}
```

My Technology Skill

Technology	My level	Experience	I Use It For
C++	Intermediate	2 years	Learning OOP and DSA Concepts in deep
Python	Intermediate	1 year	Making Real-World small Projects
MS-WORD	Master	8 years	Making Assigments , Resume`, sometimes for Documentations
MS-Excel	Intermediate	5 years	Keeping Records, Cleaning Data, small Data-bases
SQL	Beginner	2 months	Just Exploring to Work Big Data-Bases
html & CSS	Intermediate	1 year	Making portfolio, Sometimes for Documentation
Git & GitHub	Intermediate	1 month	Code Organization and Version History, Project Copies
Markdown	Master	8 months	Clean Documentation, Readable Notes
Prompt-Engineering	Above Intermediate	1.5 years	Making small projects, Exploration of other Fields, Learning something *NEW*

My Learning Journey

1. Started learning Basics of WORD and EXCEL
2. Started learning C-language
3. Learned concepts of ***Functions, Analytical-geometry, Matrices***
4. Started learning ***pointers, functions, loops, file-Handling***
5. Started Python language
6. learned
7. Data Structures and Object-Oriented Programming in Python
8. Started C++ , learned Basics
9. Object-Oriented Programming, File-Handling in C++ , Memory Management
10. Learned Discrete Mathematics,
11. Started Data Structures in C++
12. Started learning *Core Concepts* of ***Linear Algebra, Calculus, Assembly Language***

My Learning Areas

- **Programming**

    - C
        - Functions
        - Pointers
        - Loops
        - File Handling
    - Python
        - Data Structures
        - Object-Oriented Programming
        - Inheritance and Its types
    - C++
        - Basics
        - Object-Oriented Programming (*polymorphism, Virtual Classes*)
        - File Handling
        - Memory Management
        - Data Structures
- **Computer & Office Skills**
    - Microsoft Word
    - Microsoft Excel
    - Microsoft PowerPoint
- **Mathematics**
    - Analytical Geometry
    - Matrices
    - Discrete Mathematics
    - Linear Algebra
    - Calculus
- **Computer Science Fundamentals**
    - Data Structures
    - Object-Oriented Programming
    - Memory Management
    - Binary File Handling
- **Low-Level Computing**
    - Assembly Language

- Pointers
- Memory Management

My Favourite Technologies

1. Python

Why I like it: I like Python because it is beginner-friendly, versatile, and easy to understand. It can be used for many different areas, including automation, data science, web development, and artificial intelligence. I also like its simple syntax because it allows me to focus more on solving problems rather than writing complicated code.

Where I have used it: I have used Python while learning programming fundamentals, data structures, and object-oriented programming.

Useful resource: Python Official Documentation

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2. C++

Why I like it: I find C++ interesting because it gives me a tighter grip on memory management and how programs work internally. Learning pointers, memory allocation, and object-oriented programming has helped me understand what happens behind the scenes when a program runs.

Where I have used it: I have used C++ for learning data structures, object-oriented programming, file handling, and memory management.

Useful resource: C++ Reference

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3. Google Drive

Why I like it: I like cloud services such as Google Drive because they make it easy to store, organize, and access files from different devices. They also reduce the need to depend completely on local storage and make sharing files much easier.

Where I have used it: I have used Google Drive for storing study materials, documents, assignments, and other learning resources.

Useful resource: Google Drive

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4. Zapier

Why I like it: I find Zapier interesting because it shows how different applications can work together through automation. It helped me understand the idea that repetitive tasks can be connected and automated without having to manually perform every step.

Where I have used it: I have explored Zapier to understand workflow automation, Making automated (**E-mail, Forms, Excel**) Reply and Organization, and how different online services can communicate with each other.

Useful resource: Zapier

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5. Git & GitHub

Why I like it: I like Git and GitHub because they make it easier to manage my code and keep track of changes. Git helps me understand version control, while GitHub allows me to store, organize, and share my projects online. They are also useful for collaborating with other developers.

Where I have used it: I have used Git and GitHub to manage my programming projects, save different versions of my code, and organize my learning work.

Useful resource: GitHub Docs

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6. LDPlayer

Why I like it: I find LDPlayer interesting because it allows Android applications and games to run on a computer. It is also interesting from a technical perspective because it demonstrates how an emulator creates an environment where software designed for one platform can operate on another.

Where I have used it: I have used LDPlayer to run and explore Android applications and games on my computer.

Useful resource: LDPlayer Official Website

Project Checklist

Implemented Tasks

- [x] Selected the project topic: **Universal Media Downloader**
- [x] Set up the Python project environment
- [x] Integrated `yt-dlp` for video extraction and downloading
- [x] Added support for downloading images and GIFs
- [x] Added retry and fallback mechanisms using `requests` and `curl`

- [x] Added optional `ffmpeg` / `ffprobe` support
- [x] Implemented media probing and container-fixing functionality
- [x] Added support for single, multiple, and batch URL downloads
- [x] Added configurable video and image output directories
- [x] Added cookie-file support
- [x] Implemented JSONL and human-readable download logs
- [x] Added error handling and verbose logs for failed downloads
- [x] Added environment-based configuration
- [x] Documented installation, usage, configuration, and troubleshooting

Planned Tasks

- [ ] Add a **GUI (Graphical User Interface)** for easier use
- [ ] Improve download speed and overall performance
- [ ] Add more powerful downloading and processing features
- [ ] Add command-line arguments using `argparse`
- [ ] Add automated unit tests
- [ ] Refactor the downloader into a reusable importable module
- [ ] Improve media-source compatibility and detection

Mathematical Expressions

The following mathematical equations are used in my **Black Hole Simulation Project**:

Simulation of light rays moving through the curved spacetime around a Schwarzschild black hole. Rays are launched from configurable emitter presets, integrated with a Runge–Kutta 4 (RK4) solver, and rendered in real time with impact-parameter-based coloring, fading motion trails, and reference circles for the event horizon, photon sphere, and ISCO.

1. Schwarzschild Radius

$$r_s = \frac{2GM}{c^2}$$

This equation is used to calculate the **Schwarzschild radius** of the black hole.

2. Critical Impact Parameter

$$b_{\text{crit}} = \frac{3\ \ 3}{2} r_s$$

This equation is used in the simulation to calculate the **critical impact parameter** for the light rays.

3. Radial Motion of Light

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 - \left(1 - \frac{r_s}{r}\right)\frac{L^2}{r^2}$$

This equation is used to determine the **radial motion of the light rays** around the black hole.

4. Angular Motion

$$\frac{d\phi}{d\lambda} = \frac{L}{r^2}$$

This equation is used to calculate the **angular movement of the light rays**.

Quotes

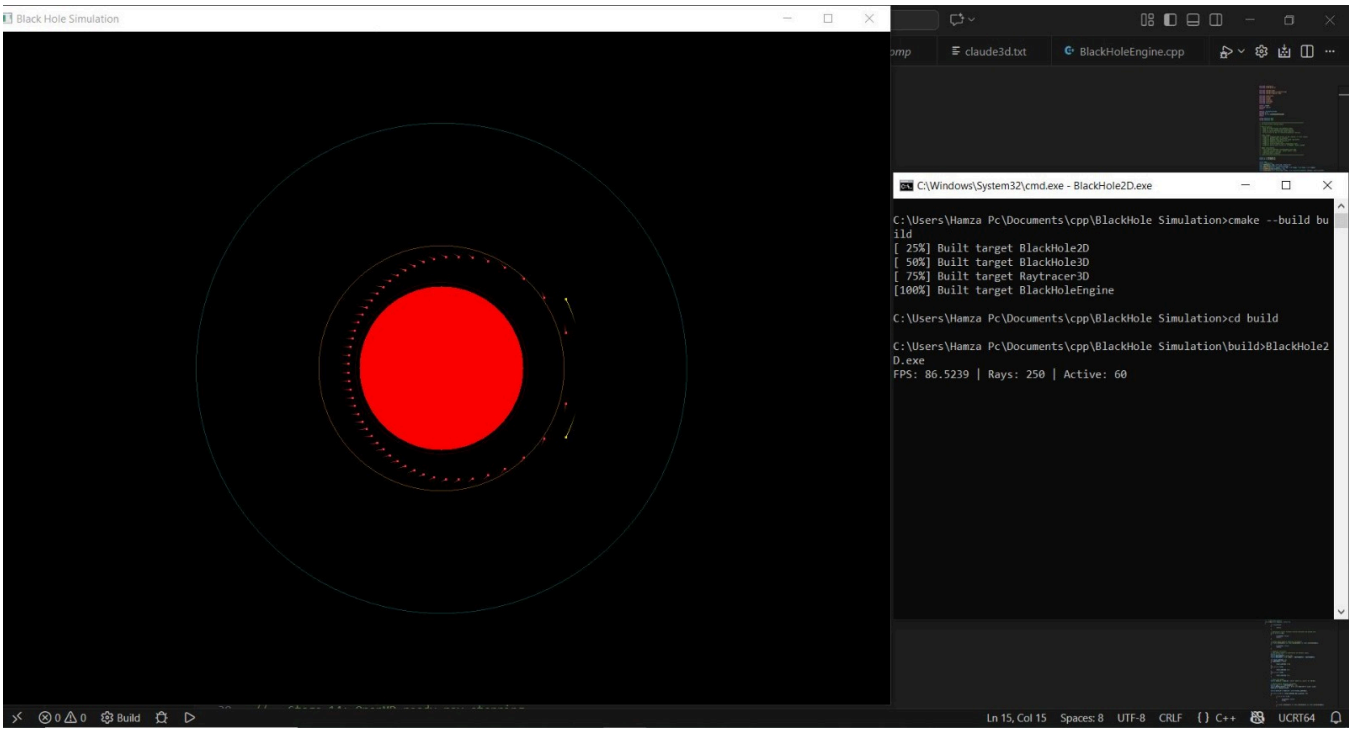
> “The important thing is to never stop questioning.”
>
> — Albert Einstein

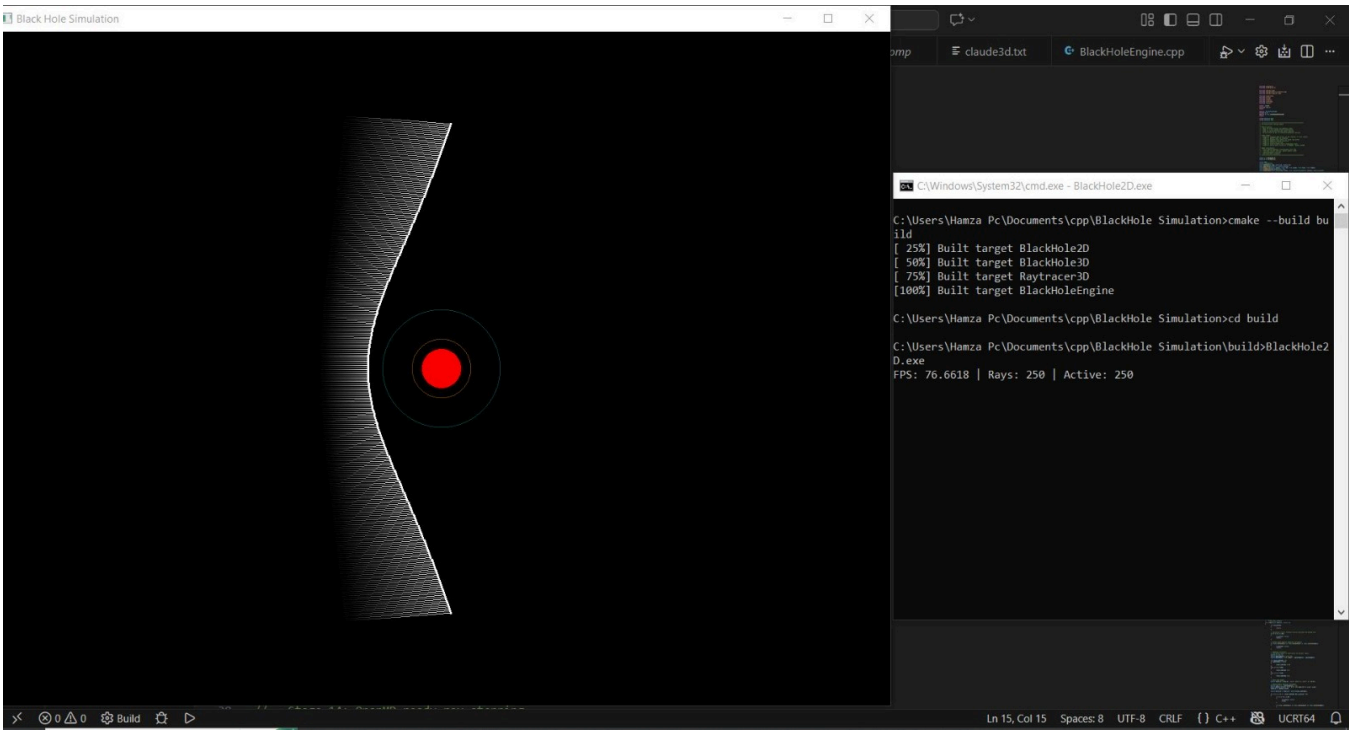
> “Knowledge is better than wealth; knowledge protects you, while you have to guard wealth.”
>
> — Hazrat Ali (RA)

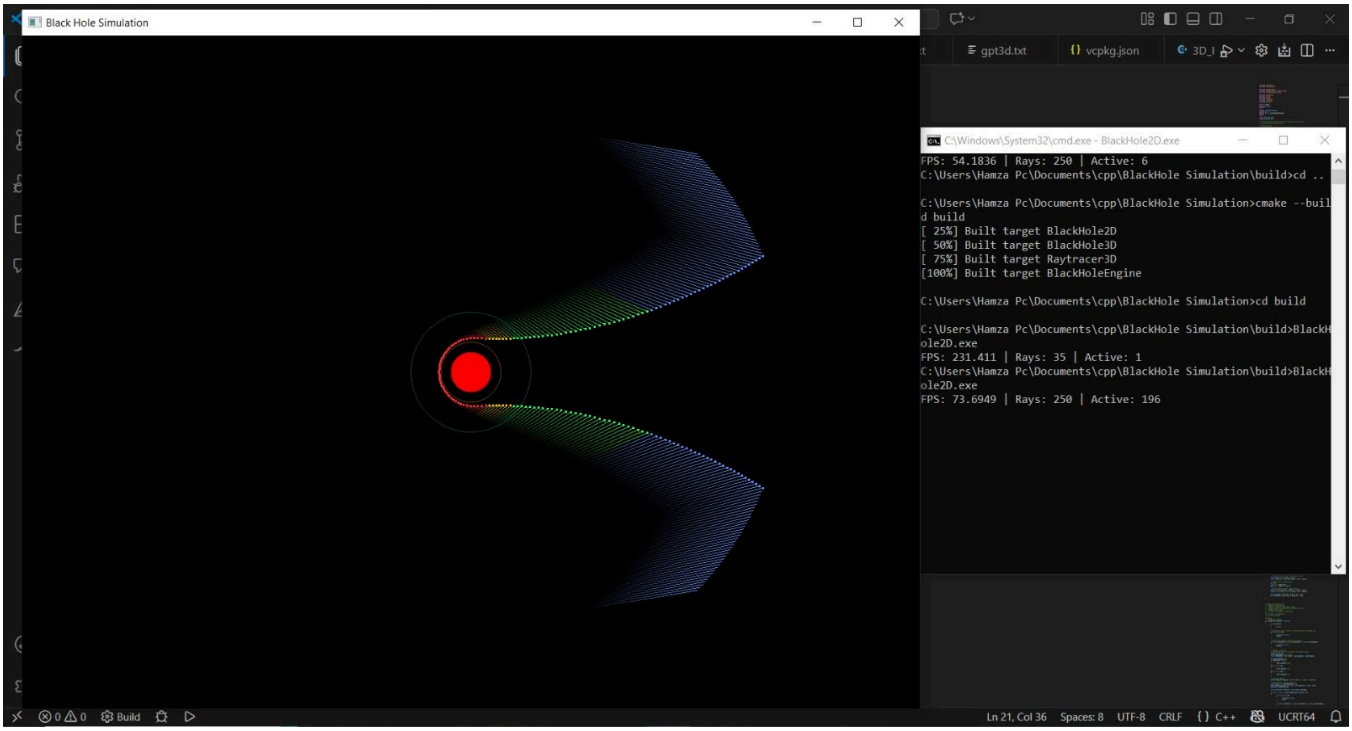
Some Pictures of Black-Hole Simulation

The following images are from *Black-Hole Simulation Project :*

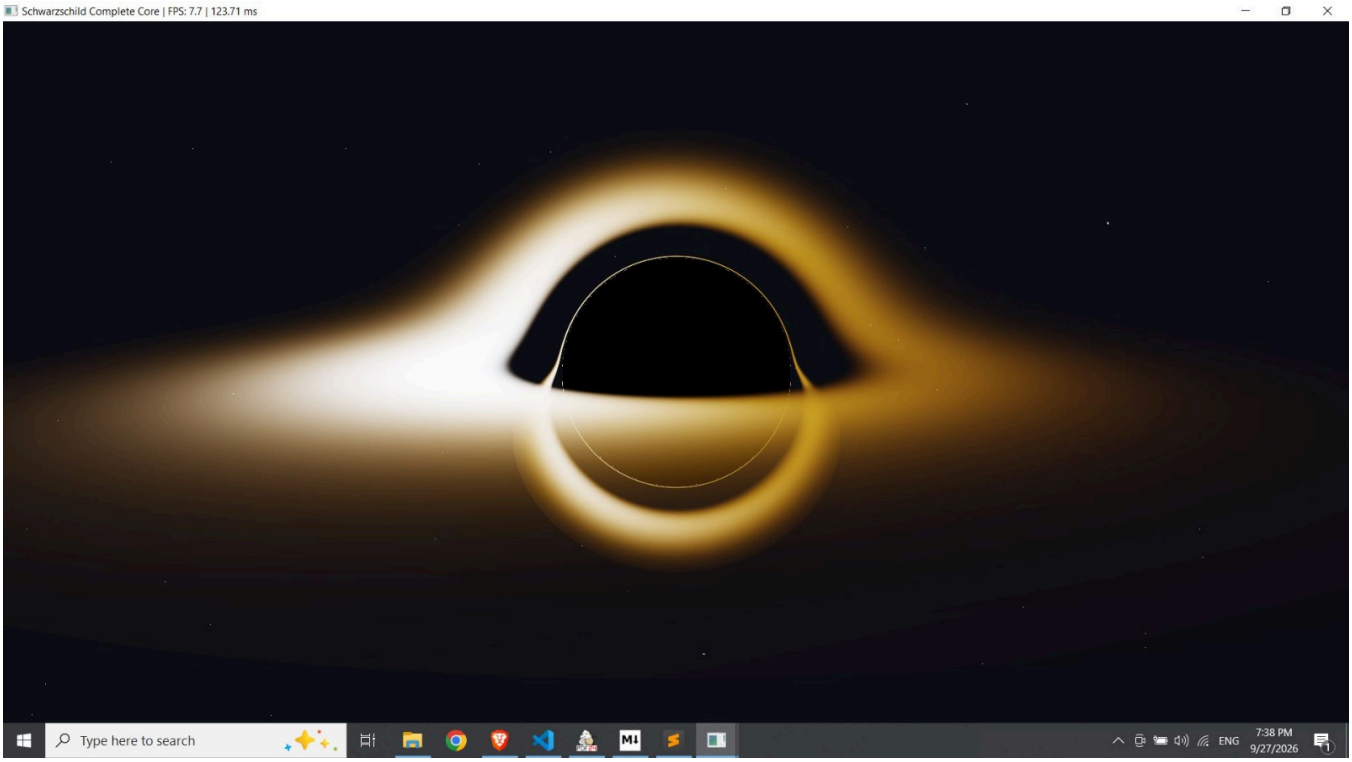
Two Dimensional (2D)

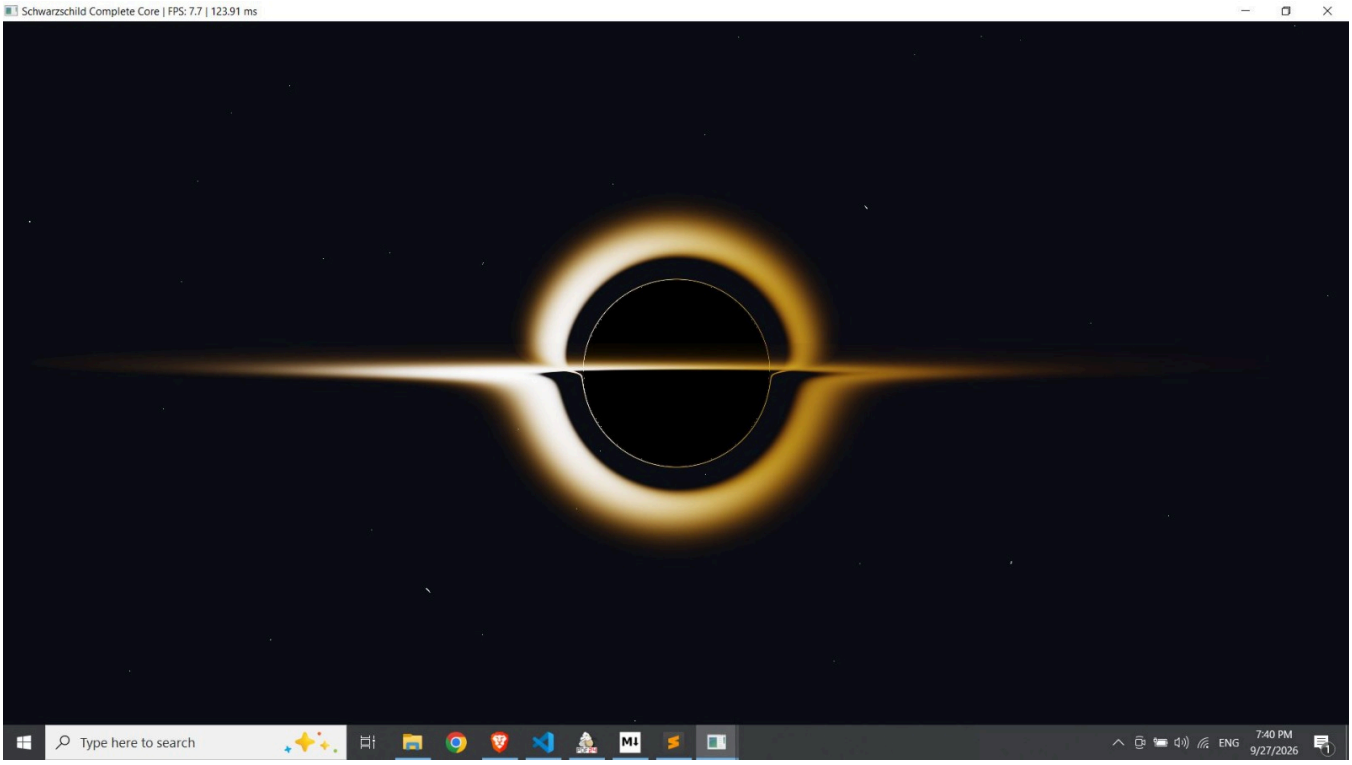


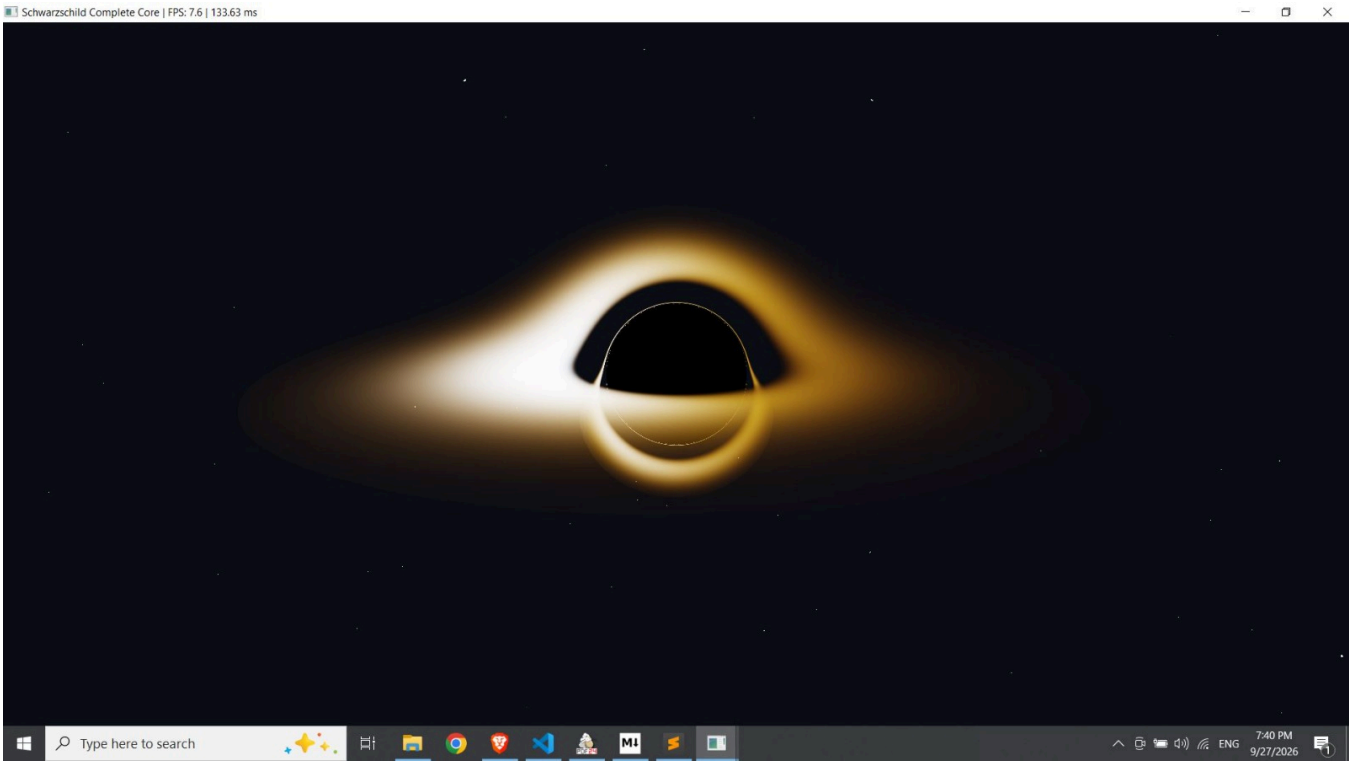



Black Hole Simulation
gpt3d.txt
vcpkg.json
C:\Windows\System32\cmd.exe - BlackHole2D.exe
FPS: 54.1836 | Rays: 250 | Active: 6
C:\Users\Hamza Pc\Documents\cpp\BlackHole Simulation\build>cd ..
C:\Users\Hamza Pc\Documents\cpp\BlackHole Simulation>cmake --build build
[25%] Built target BlackHole2D
[50%] Built target BlackHole3D
[75%] Built target Raytracer3D
[100%] Built target BlackHoleEngine
C:\Users\Hamza Pc\Documents\cpp\BlackHole Simulation>cd build
C:\Users\Hamza Pc\Documents\cpp\BlackHole Simulation\build>BlackHole2D.exe
FPS: 231.411 | Rays: 35 | Active: 1
C:\Users\Hamza Pc\Documents\cpp\BlackHole Simulation\build>BlackHole2D.exe
FPS: 73.6949 | Rays: 250 | Active: 196
Build
Ln 21, Col 36
Spaces: 8
UTF-8
CRLF
C++
UCRT64

Three Dimensional (3D)







Markdown Features I Practiced

- [x] Headings
- [x] Bold
- [x] Italic
- [x] Ordered List
- [x] Unordered List
- [x] Nested List
- [x] Links
- [x] Images
- [x] Tables
- [x] Blockquotes
- [x] Mathematical Expressions
- [x] Horizontal Rule